



## DO-IT-YOURSELF: CIRCUIT IDEA

diode ZD2 is more than 5.6V. So, transistor T2 is in conduction mode. Consequently, relay RL1 energises and electrical appliance connected at its contacts gets the AC power supply.

When mains voltage goes high, say more than 230V AC, transistor T1 conducts because voltage at cathode of Zener diode ZD1 becomes greater than 5.6V, and transistor T2 does not conduct. As a result, relay RL1 de-energises and electrical appliance gets disconnected from the AC mains power supply to protect the appliance.

If the mains voltage becomes low, say less than 190V AC, transistor T1 does not conduct because voltage at cathode of Zener diode ZD1 is less than 5.6V. So, transistor T2 also does not conduct. As a result, relay RL1 de-energises and electrical appliance

once again gets disconnected from the mains power supply.

To avoid chattering of relay RL1, especially during high/low voltage threshold level settings, capacitor C2 is connected across the relay coil.

### Construction and testing

An actual-size PCB layout for the circuit is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic box.

Connect the electrical appliance to connector J2 and 230V AC mains to connector/socket J1. The AC mains socket should be fixed at the back side of the cabinet to connect the electrical appliance.

Connect LED1 and potmeters VR1 and VR2 on the front of the cabinet. VR1 and VR2 are used

### PARTS LIST

#### Semiconductors:

- T1, T2 - 2N2222 NPN transistor
- D1-D3 - 1N4007 rectifier diode
- LED1 - 5mm LED
- ZD1, ZD2 - 5.6V Zener diode

#### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R3 - 1-kilo-ohm
- VR1, VR2 - 4.7-kilo-ohm potmeter

#### Capacitors:

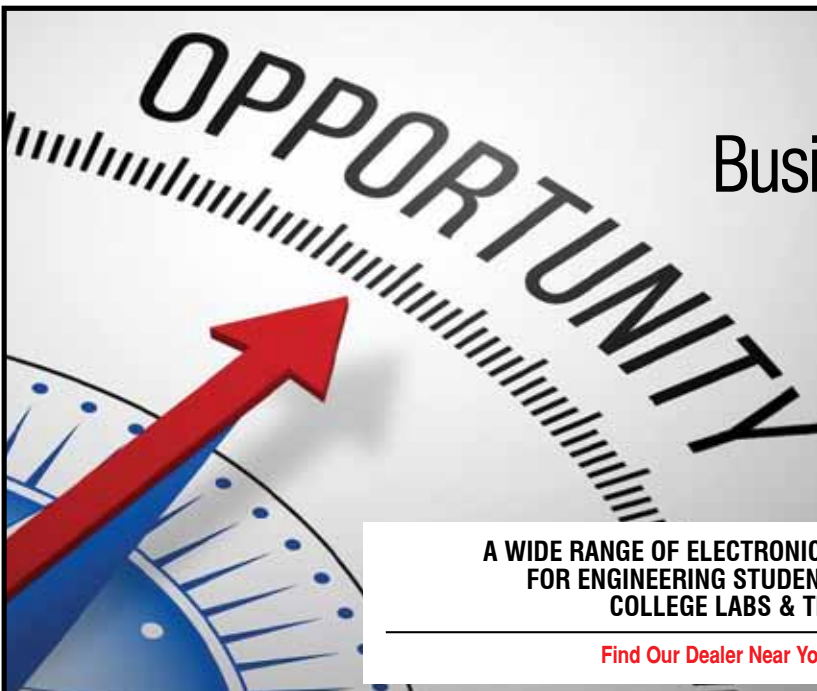
- C1 - 220 $\mu$ F, 35V electrolytic
- C2 - 470 $\mu$ F, 35V electrolytic

#### Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 500mA secondary transformer
- EA - AC mains operated appliance
- J1, J2 - 2-pin terminal connector
- RL1 - 12V SPDT Finder relay

to set the low and high voltages, respectively, to trigger the circuit for protection of appliances. **EFY**

S.C. Dwivedi is an electronics enthusiast, circuit designer and technical article writer



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